LOGIC BOMB

Defusal Manual · v2.1

> Check numbered rules **top to bottom**. Use **only the first TRUE rule**.

THE CASING (DEFUSER ROTATES THE BOMB TO READ IT)

ITEM	WRITE DOWN
Serial number	**D** = set of its digits · **V** = set of its vowels
Batteries	**b** = number of batteries
Indicators	Is any indicator **lit**?

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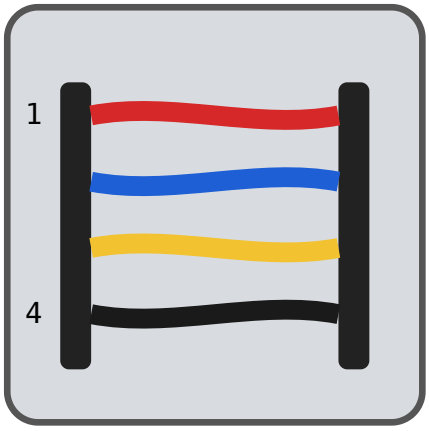
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1 Wires



1. Count the wires (3–6).
2. Wire 1 is the **top** wire. “Last” = lowest.
3. Cut **one** wire.

R(w) red · **B(w)** blue · **Y(w)** yellow · **K(w)** black · **W(w)** white

3 WIRES

1	¬∃w R(w)	**cut wire 2**
2	∀w (R(w) ∨ B(w))	**cut last wire**
3	∃w W(w)	**cut last white**
4	otherwise	**cut wire 1**

4 WIRES

1	∀w ¬Y(w)	**cut wire 1**
2	∃w K(w)	**cut last black**
3	∀w (R(w) ∨ Y(w))	**cut wire 3**
4	otherwise	**cut wire 4**

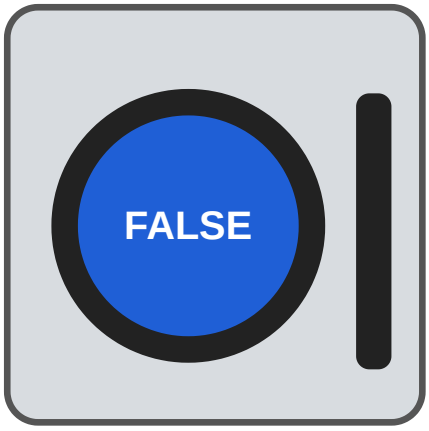
5 WIRES

1	∃w K(w) ∧ ∃w W(w)	**cut wire 4**
2	∀w ¬R(w)	**cut wire 1**
3	∃d ∈ D (d is odd)	**cut last wire**
4	otherwise	**cut wire 2**

6 WIRES

1	∀w ¬Y(w)	**cut wire 3**
2	∃w W(w)	**cut last white**
3	∃w ∃v (w ≠ v ∧ R(w) ∧ R(v))	**cut last red**
4	otherwise	**cut wire 6**

2 The Button



1. Find the first TRUE rule.
2. **TAP** = press and let go.
3. **HOLD** = keep holding; a strip lights up. Release using the table below.

p	button is red
q	label says TRUE
r	2 or more batteries
s	an indicator is lit

RULES

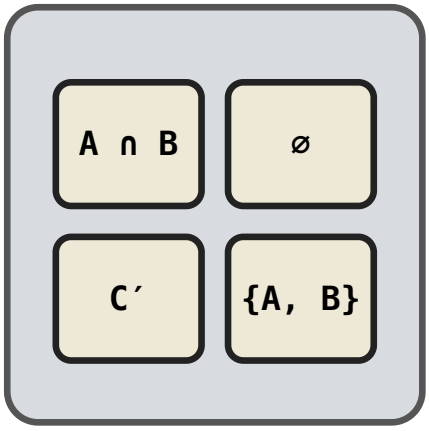
1	**p ∧ q**	**TAP**
2	**¬p ∧ ¬r**	**HOLD**
3	**q ∨ s**	**TAP**
4	otherwise	**HOLD**

RELEASE WHEN THE TIMER SHOWS…

STRIP	THE DIGIT IN
Blue	**{1,2,3,4} ∩ {4,5,6}**
White	**{1,3,5} − {3,5}**
Yellow	**{5,6,7} ∩ {1,3,5}**
Red	**{x ∈ ℤ | 1 < x < 3}**

The digit can be anywhere on the timer.

3 Set Keypad



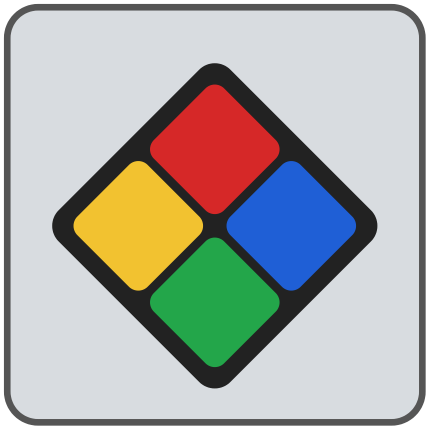
1. Each key shows a set.
2. Find each set’s size **|S|**.
3. Press keys from **smallest** to **largest**.

A = {1, 2, 3, 4} · **B = {3, 4, 5}** · **C = {2, 4, 6, 8}**
Universe **𝕌 = {1, …, 8}** (for complements like **A′**)

WORKSHEET

KEY SHOWS	ELEMENTS	SIZE	ORDER

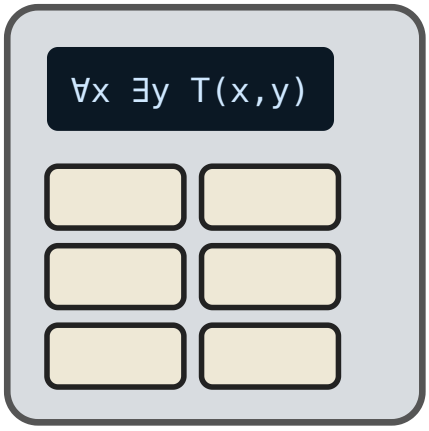
4 Simon Maps



1. Colors flash in a sequence that grows each stage.
2. For each flash **x**, press **f(x)**, in order.
3. Repeat the whole sequence every stage.

SERIAL	FUNCTION
has a vowel **(V ≠ ∅)**	**f = {(Red, Blue), (Blue, Red), (Green, Yellow), (Yellow, Green)}**
no vowel **(V = ∅)**	**f = {(Red, Yellow), (Blue, Green), (Green, Blue), (Yellow, Red)}**

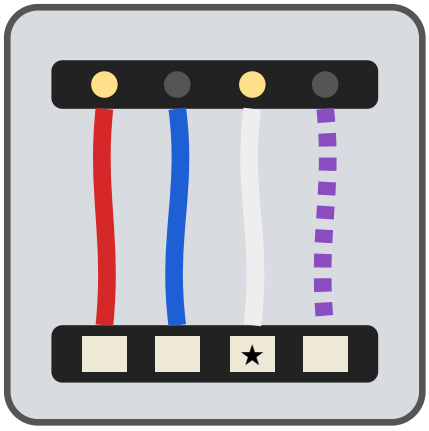
5 Who’s on ∀irst



1. **`T(x, y)`** = “x is taller than y”. Domain: the class.
2. Translate the screen into English.
3. Press that sentence’s code word. 3 stages.

SENTENCE	CODE WORD
Someone is taller than everyone.	**COMET**
Everyone has someone taller than them.	**EMBER**
No one is taller than anyone.	**GLACIER**
Everyone is taller than someone.	**BISHOP**
Someone is taller than someone.	**DYNAMO**
Someone has everyone taller than them.	**FALCON**
Everyone is taller than everyone.	**ATLAS**
Someone is taller than no one.	**HARBOR**

6 Venn Wires



1. Wires are numbered **left to right**.
2. For each wire: color, LED on/off, ★ or blank.
3. Cut every wire in K. Order doesn’t matter.

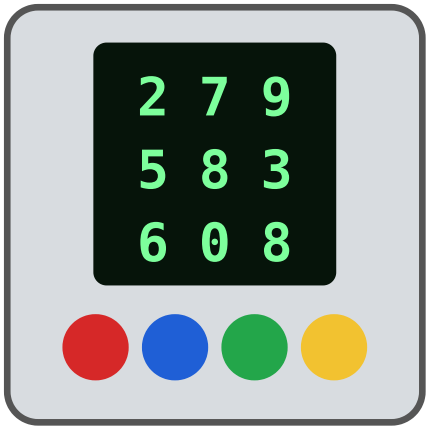
R	red or striped
B	blue or striped
L	LED on
S	★ below

$$K = (R \cap L) \cup (S - B)$$

WORKSHEET

#	R	B	L	S	CUT?
1					
2					
3					
4					
5					
6					

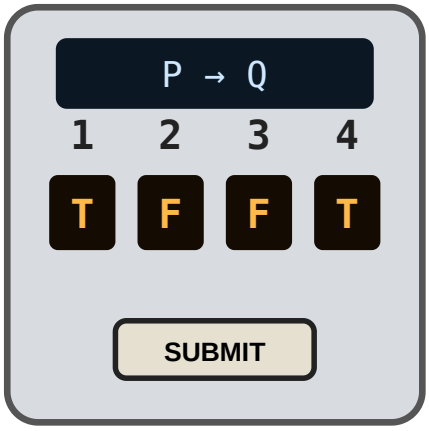
7 Quantifier Grid



1. Rows **r**: top to bottom. Columns **c**: left to right.
2. **x** = the digit at row r, column c.
3. Press the color of the first TRUE rule.

1	`∃r ∀c (x is even)`	**RED**
2	`∀r ∃c (x = 0)`	**BLUE**
3	`∃c ∀r (x > 5)`	**GREEN**
4	otherwise	**YELLOW**

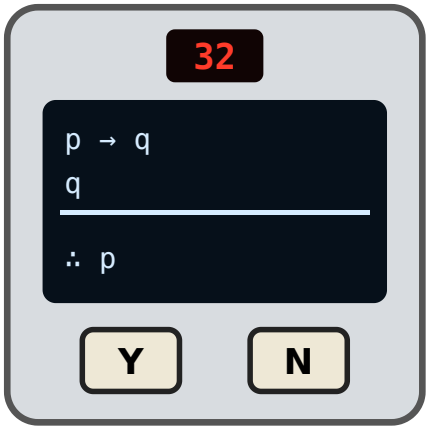
8 Truth Table



1. Set each window to the formula’s value for its row.
2. Arrows flip T ↔ F.
3. Press **SUBMIT**.

WINDOW	P	Q	FORMULA
1	T	T	
2	T	F	
3	F	T	
4	F	F	

N Argument Check (needy)



1. Wakes up now and then with a 40-second countdown.
2. **Y** = valid · **N** = invalid.
3. Never disarms. Keep answering it.

VALID → Y
p → q, p ∴ q
p → q, ¬q ∴ ¬p
p → q, q → r ∴ p → r
p ∨ q, ¬p ∴ q
p ∴ p ∨ q
p ∧ q ∴ p
∀x P(x) ∴ P(c)
P(c) ∴ ∃x P(x)
∀x (P(x) → Q(x)), P(c) ∴ Q(c)
¬∀x P(x) ∴ ∃x ¬P(x)

INVALID → N
p → q, q ∴ p
p → q, ¬p ∴ ¬q
p ∨ q, p ∴ ¬q
∀x (P(x) → Q(x)), Q(c) ∴ P(c)
∃x P(x) ∴ ∀x P(x)
¬∀x P(x) ∴ ∀x ¬P(x)

✓ Answer Key

Casing for every example: serial **KA4T74** (**D = {4, 7}**, **V = {A}**), 2 batteries, NEG lit.

MODULE	BOMB SHOWS	WORK	ANSWER
Wires	red, yellow, black, blue	Rule 1 F (yellow). Rule 2: a black exists.	**cut wire 3**
Wires	red, yellow, blue, black, blue	Rule 1 F (no white). Rule 2 F. Rule 3: 7 is odd.	**cut wire 5**
Button	blue, “NOT”	p F, q F, r T, s T. Rule 2: T ∧ F = F. Rule 3: F ∨ T = T.	**TAP**
Button	held, yellow strip	**{5,6,7} ∩ {1,3,5} = {5}**	**release on 5**
Keypad	A ∪ B, ∅, C − B, B ∩ C	sizes 5, 0, 3, 1	**∅, B∩C, C−B, A∪B**
Simon	red; then red, green	V ≠ ∅ → first function	**blue; blue, yellow**
∀irst	**∀x ∃y T(x,y)**	Everyone is taller than someone.	**BISHOP**
∀irst	**∃y ∀x T(x,y)**	Someone has everyone taller than them.	**FALCON**
Venn	1 red/LED · 2 blue/LED/★ · 3 white/★ · 4 striped/LED · 5 red/★ · 6 blue	**R∩L = {1,4}, S−B = {3,5}**	**cut 1, 3, 4, 5**
Grid	7 1 3 / 8 2 1 / 9 5 1	Column 1 = 7, 8, 9: all > 5.	**GREEN**
Truth	**¬P → Q**	TT: T · TF: T · FT: T · FF: F	**T T T F**
Argument	**∀x (P(x) → Q(x)), Q(c) ∴ P(c)**	Affirming the conclusion.	**N**